



## PCF lab Certificate Rotation Estimate

- **Director:** 192.168.2.2
- **Generated:** 2026-06-10 11:19:48 UTC
- **Horizon:** all certificates (no expiry limit)
- **Topology source:** maestro tp
- **Update rules:** bosh manifest (per-IG max\_in\_flight/canaries)

### Summary

| Metric                          | Value                  |
|---------------------------------|------------------------|
| Expiring certs (all (no limit)) | 112 (✗ 0 · ⚠ 0)        |
| — leaf / CA                     | 105 / 7                |
| Live VMs (DB nodes)             | 7 (2)                  |
| One foundation-wide Apply       | 1h 00m – 1h 50m        |
| <b>Estimated rotation time</b>  | <b>3h 00m – 5h 30m</b> |

### Foundation inventory

| Deployment              | Type | VMs | Est. cert-rotation time |
|-------------------------|------|-----|-------------------------|
| cf-dbe1a7580979a87638e7 | cf   | 7   | 3h 00m – 5h 30m         |

2 instance group(s) have serial:false — they may update in parallel, so real time can come in under the (conservative serial) estimate.

## CA rotations (3-phase)

| Certificate                                                                              | Expires              |
|------------------------------------------------------------------------------------------|----------------------|
| <code>.properties.nats_client_ca.3506e33356bef86af963</code>                             | 2030-05-29T03:06:57Z |
| <code>.properties.root_ca.3506e33356bef86af963</code>                                    | 2030-05-29T03:06:57Z |
| <code>/p-bosh/cf-dbe1a7580979a87638e7/diego-instance-identity-intermediate-ca-2-7</code> | 2028-06-09T04:19:47Z |
| <code>/opsmgr/bosh_dns/tls_ca</code>                                                     | 2030-05-29T16:32:50Z |
| <code>/cf/diego-instance-identity-root-ca-2-6</code>                                     | 2029-05-29T16:32:45Z |
| <code>/services/tls_ca</code>                                                            | 2031-05-29T03:24:34Z |
| <code>opsman-root-ca:3506e333</code>                                                     | 2030-05-29T03:06:57Z |

## Leaf certificates

| Deployment                           | Leaf certs | Apply                       |
|--------------------------------------|------------|-----------------------------|
| <code>p-bosh (not VM-counted)</code> | 11         | folded into shared apply #2 |
| <code>cf-dbe1a7580979a87638e7</code> | 94         | folded into shared apply #2 |

## Estimate breakdown

| Campaign              | Applies                                               | Time (low – high) |
|-----------------------|-------------------------------------------------------|-------------------|
| Leaf certs            | folded into the shared apply #2                       | included          |
| CA certs (full batch) | 3× foundation-wide on all tiles —<br>all CAs + leaves | 3h 00m – 5h 30m   |
| Total                 |                                                       | 3h 00m – 5h 30m   |

## Model & assumptions

- 20m overhead per Apply Changes; stateless 4–10m/VM, DB 10–20m/node.
- A FOUNDATION CA = 3 foundation-wide applies; a services TLS CA = 2 foundation + 1 deployment applies; a deployment CA = 3 on its deployment.

- Estimate is **Apply Changes compute time only** — it excludes change-window / approval gaps between phases, which often dominate the wall-clock for CA rotations.